12/12

MATHEMATICS 2210

Quiz 5, Fall semester 2012

Name: KEY

1. (3 pts) Find the equation of the tangent plane to

$$z = f(x, y) = x^2/y \qquad \text{at} \qquad p = (1, 2).$$

$f(1,2) = 1/2$

$g(x,y,z) = \frac{x^2}{y} - z$ $\qquad \nabla g = \left(\frac{2x}{y}, -\frac{x^2}{y^2}, -1\right)$

$\nabla g(1,2,1/2) = \left(1, -\frac{1}{4}, -1\right)$

$0 = \left(1, -\frac{1}{4}, -1\right)\left(x-1, y-2, z-\frac{1}{2}\right) = x-1-\frac{1}{4}(y-2)-\left(z-\frac{1}{2}\right)$

or $\quad x - \frac{y}{4} - z = 0$

2. (3 pts) Show that the following limit does not exist.

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{(x+y)^2}$$

$y = 0 \quad \lim_{x\to 0} \frac{x^2}{x^2} = 1$

$x = y \quad \lim_{x\to 0} \frac{2x^2}{(2x)^2} = \frac{1}{2}$

DO NOT MATCH

SO LIMIT DOES NOT EXIST

3. (3 pts) Show that

$$\lim_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2} = 0.$$

(Hint: use the Squeeze Theorem.)

$$\underset{\downarrow \atop 0}{x^2-y^2} \le \frac{x^4-y^4}{x^2+y^2} \le \frac{x^4+y^4}{x^2+y^2} \le \frac{(x^2+y^2)^2}{x^2+y^2} = \underset{\searrow \atop 0}{x^2+y^2}$$

Since $\lim\limits_{(x,y)\to(0,0)} x^2-y^2 = 0 = \lim\limits_{(x,y)\to(0,0)} x^2+y^2$

By the squeeze theorem, $\lim\limits_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2} = 0$

4. (3 pts) Compute

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2) - (x^2+y^2)}{(x^2+y^2)^3}.$$

Let $t = x^2+y^2$ then we must compute

$$\lim_{t\to 0} \frac{\sin t - t}{t^3} \overset{\text{l'Hopital}}{=} \lim_{t\to 0} \frac{\cos t - 1}{3t^2} = \lim_{t\to 0} \frac{-\sin t}{6t}$$

$$= \lim_{t\to 0} -\frac{\cos t}{6} = -\frac{1}{6}$$